Question 6: Implement Kubernetes deployment strategies such as rolling updates and rollbacks, and demonstrate scaling of the application replicas.

Course outcome: CO1, CO2

A rolling update from image 2.0.0 to 2.0.1 was performed and observed, the change was rolled back, and the deployment was scaled from 2 to 4 replicas and back.

## Rolling update

```
$ docker build --build-arg APP_VERSION=2.0.1 -f docker/Dockerfile -t devops-lab2-app:2.0.1 .
$ minikube image load devops-lab2-app:2.0.1
$ kubectl set image deployment/lab2-app lab2-app=devops-lab2-app:2.0.1
deployment.apps/lab2-app image updated

pod transitions captured during the rollout:
-- t=5  1/1 Running (new)   1/1 Running (new)   Terminating (old)
-- t=8  1/1 Running (new)   1/1 Running (new)   Terminating (old) x2

$ curl http://192.168.49.2:30080/health
{"status":"UP","version":"2.0.1"}
```

## Rollback

```
$ kubectl rollout undo deployment/lab2-app
deployment.apps/lab2-app rolled back
$ kubectl get pods -l app=lab2-app -o jsonpath image field
lab2-app-88f4c9fc9-bv4qd:  devops-lab2-app:2.0.0
lab2-app-88f4c9fc9-snrr5:  devops-lab2-app:2.0.0

$ curl http://192.168.49.2:30080/health   (after old pods terminated)
{"status":"UP","version":"2.0.0"}
```

## Scaling

```
$ kubectl scale deployment/lab2-app --replicas=4
deployment.apps/lab2-app scaled

$ kubectl get pods -l app=lab2-app
4 pods Running
$ kubectl get deployment lab2-app
lab2-app  4/4  4  4

$ kubectl scale deployment/lab2-app --replicas=2
$ kubectl get pods -l app=lab2-app
2 pods Running
```

```
abhinav@LAPTOP-BSV7PCJK: ~/devops-lab1-ci  —  terminal
lab2-app-56dd487c9b-682tp   1/1     Running       0          25s
lab2-app-88f4c9fc9-9fjc5    1/1     Terminating   0          77s
lab2-app-88f4c9fc9-kdrsn    1/1     Terminating   0          65s
abhinav@LAPTOP-BSV7PCJK:~$ curl health after rolling update:
{"status":"UP","version":"2.0.1"}
abhinav@LAPTOP-BSV7PCJK:~$ kubectl rollout history deployment/lab2-app
deployment.apps/lab2-app
REVISION  CHANGE-CAUSE
1         <none>
2         <none>
3         <none>
4         <none>
### ROLLBACK: undo to the previous revision (2.0.0 image)
abhinav@LAPTOP-BSV7PCJK:~$ kubectl rollout undo deployment/lab2-app
Warning: resource deployments/lab2-app was previously managed with 'kubectl apply'. Rolling back will not
deployment.apps/lab2-app rolled back
Waiting for deployment "lab2-app" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "lab2-app" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "lab2-app" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "lab2-app" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "lab2-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "lab2-app" rollout to finish: 1 old replicas are pending termination...
deployment "lab2-app" successfully rolled out
abhinav@LAPTOP-BSV7PCJK:~$ kubectl get pods -l app=lab2-app
NAME                        READY   STATUS        RESTARTS   AGE
lab2-app-56dd487c9b-2ctj2   1/1     Terminating   0          59s
lab2-app-56dd487c9b-682tp   1/1     Terminating   0          72s
lab2-app-88f4c9fc9-bv4qd    1/1     Running       0          12s
lab2-app-88f4c9fc9-snrr5    1/1     Running       0          24s
abhinav@LAPTOP-BSV7PCJK:~$ curl health after rollback:
{"status":"UP","version":"2.0.1"}
--> the app is serving 2.0.0 again after the rollback
abhinav@LAPTOP-BSV7PCJK:~$ kubectl rollout history deployment/lab2-app
deployment.apps/lab2-app
REVISION  CHANGE-CAUSE
1         <none>
2         <none>
4         <none>
```

Fig: Rolling update: new pods Running while old pods terminate

```
abhinav@LAPTOP-BSV7PCJK: ~/devops-lab1-ci  —  terminal
5         <none>
### Post-rollback verification (after old pods fully terminated)
abhinav@LAPTOP-BSV7PCJK:~$ kubectl get pods -l app=lab2-app -o jsonpath image field
lab2-app-88f4c9fc9-bv4qd: devops-lab2-app:2.0.0
lab2-app-88f4c9fc9-snrr5: devops-lab2-app:2.0.0
abhinav@LAPTOP-BSV7PCJK:~$ curl http://192.168.49.2:30080/health
{"status":"UP","version":"2.0.0"}
# OBSERVATION: the curl taken immediately after 'rollout undo' returned 2.0.1
# because the terminating 2.0.1 pods were still registered in the service
# endpoints for a few seconds. Once they finished terminating, the service
# served only 2.0.0 pods. Zero-downtime is preserved during this overlap.
```

Fig: Rollback verified: 2.0.0 image running again

```
abhinav@LAPTOP-BSV7PCJK: ~/devops-lab1-ci  —  terminal
### SCALING: increase replicas from 2 to 4
abhinav@LAPTOP-BSV7PCJK:~$ kubectl scale deployment/lab2-app --replicas=4
deployment.apps/lab2-app scaled
abhinav@LAPTOP-BSV7PCJK:~$ kubectl get pods -l app=lab2-app
NAME                        READY   STATUS        RESTARTS   AGE
lab2-app-56dd487c9b-2ctj2   1/1     Terminating   0          73s
lab2-app-56dd487c9b-682tp   1/1     Terminating   0          86s
lab2-app-88f4c9fc9-bv4qd    1/1     Running       0          26s
lab2-app-88f4c9fc9-snrr5    1/1     Running       0          38s
lab2-app-88f4c9fc9-tnb9c    1/1     Running       0          12s
lab2-app-88f4c9fc9-v8vv5    1/1     Running       0          12s
abhinav@LAPTOP-BSV7PCJK:~$ kubectl get deployment lab2-app
NAME       READY   UP-TO-DATE   AVAILABLE   AGE
lab2-app   4/4     4            4           19m
### scale back down to 2
abhinav@LAPTOP-BSV7PCJK:~$ kubectl scale deployment/lab2-app --replicas=2
deployment.apps/lab2-app scaled
abhinav@LAPTOP-BSV7PCJK:~$ kubectl get pods -l app=lab2-app
NAME                        READY   STATUS        RESTARTS   AGE
lab2-app-56dd487c9b-2ctj2   1/1     Terminating   0          85s
lab2-app-88f4c9fc9-bv4qd    1/1     Running       0          38s
lab2-app-88f4c9fc9-snrr5    1/1     Running       0          50s
lab2-app-88f4c9fc9-tnb9c    1/1     Terminating   0          24s
lab2-app-88f4c9fc9-v8vv5    1/1     Terminating   0          24s
```

Fig: Scaling 2 -> 4 -> 2 replicas

## Observations

- The rolling update replaced pods one by one - the service never went down (captured transitions show new pods Running before old pods finish Terminating).
- The version reverts with the image because APP_VERSION is baked in at build time (build ARG) rather than set through the ConfigMap.
- The health curl taken immediately after `rollout undo` still returned 2.0.1 for a few seconds: the terminating pods were still registered in the service endpoints.
- Scaling is instantaneous on a single-node cluster - two extra Running pods appeared within seconds.

## Conclusion

All experiments were performed on Ubuntu 24.04 LTS with Docker, Docker Compose, Minikube and kubectl. The evidence (command logs, screenshots and commit history) is committed in the GitHub repository github.com/theabhinav0231/devops-lab-2 under the docs/ directory.